ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

ROBOTICS AND INDUSTRIAL AUTOMATION

PAPER 2 · PRACTICE B

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 4. Paper 2 style (design-related), built around one case study: a robot palletising cell. Section A: objective questions (30) · Section B: cell layout, robot selection and throughput, PLC I/O, ladder rungs, counters and robot handshaking, HMI, OEE, SCADA and Industry 4.0, safety, light-curtain muting and commissioning (70).

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INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
6. Draw diagrams neatly, using a ruler. A non-programmable calculator may be used.
7. Write neatly and legibly.

SECTION A

QUESTION 1

Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1 Palletising means …

A stacking products on a pallet in a set pattern.
B painting pallets.
C welding pallets.
D sorting products by colour.

1.2 Operators use an HMI to …

A weld parts.
B view and control the process.
C store parts.
D power the motors.

1.3 The PLC learns that a pallet is in place from a …

A sensor.
B motor.
C lamp.
D valve.

1.4 A light curtain is used to …

A light the area.
B power the PLC.
C count boxes.
D detect a person entering the danger zone.

1.5 Profinet and EtherNet/IP are …

A industrial communication standards.
B robot grippers.
C sensors.
D safety fences.

1.6 SCADA stands for …

A System Code and Data Access.
B Supervisory Control and Data Acquisition.
C Safety Control and Detection Alarm.
D Sensor Control and Drive Automation.

1.7 A vacuum gripper is best suited to …

A loose sand.
B open mesh baskets.
C sealed cardboard boxes.
D liquids.

1.8 OEE (overall equipment effectiveness) combines …

A voltage readings.
B colour checks.
C temperatures.
D availability, performance and quality.

1.9 Turning every second layer of boxes in a pallet pattern improves …

A the colour.
B the stability of the stack.
C the forklift's speed.
D the box weight.

1.10 Muting a light curtain at the pallet exit allows …

A everything to pass.
B nothing to pass.
C pallets to pass while people are still detected.
D the lights to dim.

(10 × 1) [10]

QUESTION 2

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (2.1–2.10) in the ANSWER BOOK.

2.1 A safety fence with an interlocked gate stops the robot when the gate is opened.

2.2 An HMI screen can show alarms to the operator.

2.3 The robot should restart by itself as soon as a guard door is closed.

2.4 A PLC I/O list helps electricians to wire the system correctly.

2.5 SCADA can store production data for reports.

2.6 A heavier payload allows a robot to move faster.

2.7 The cycle time sets how many boxes the cell can stack per hour.

2.8 Industry 4.0 means removing all sensors from machines.

2.9 Any visitor may reset an emergency stop.

2.10 Simulation can find collisions before the real cell is built.

(10 × 1) [10]

QUESTION 3

Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (3.1–3.10) in the ANSWER BOOK.

	COLUMN A		COLUMN B
3.1	Wooden platform on which goods are stacked	A	Profinet
3.2	Screen that operators use to monitor and control a machine	B	HMI
3.3	System that collects data from many machines in a plant	C	pallet
3.4	Detects a person entering through beams of light	D	simulation
3.5	Layout of the boxes on each layer	E	stacking pattern
3.6	Percentage that shows how effectively equipment is used	F	interlock switch
3.7	Stops the robot when a guard door is opened	G	SCADA
3.8	Testing a robot program in a 3D computer model	H	buffer conveyor
3.9	Industrial Ethernet protocol	I	light curtain
3.10	Conveyor section that holds boxes until the robot is ready	J	OEE

(10 × 1) [10]

TOTAL SECTION A: [30]

SECTION B

CASE STUDY. A juice factory wants a robot cell that stacks 8 kg boxes onto pallets. Boxes arrive on an infeed conveyor; the robot stacks 5 layers of 6 boxes on each pallet; a forklift removes full pallets. A PLC controls the cell, operators use an HMI, and production data goes to the factory's SCADA system.

QUESTION 4

4.1 Draw a neat, labelled plan view (layout) of the palletising cell showing the robot, the infeed conveyor, the pallet stations, the safety fence, the light curtain, the control cabinet and the HMI. (6)

4.2 The vacuum gripper has a mass of 4 kg. Calculate the payload needed and choose a robot rated 10 kg, 20 kg or 50 kg. Give ONE other specification that must be checked. (3)

4.3 The robot takes 6 s to place one box. Calculate the time to fill one pallet, the number of boxes stacked per hour and the number of pallets filled per hour. (4)

4.4 Explain why the cell should have two pallet stations. (3)

4.5 Choose a type of gripper for the boxes and give a reason. (2)

[18]

QUESTION 5

5.1 Draw up an I/O allocation table with at least FIVE inputs and THREE outputs for the cell. (8)

5.2 Draw the ladder rung for the infeed conveyor: it starts with the START button and keeps running (seal-in), but stops when a box reaches the pick sensor, when STOP is pressed or when the emergency stop is pressed. (5)

5.3 Explain how a PLC counter is used to tell the robot when a pallet is full and to switch to the second pallet station. (4)

5.4 Explain what handshake signals between the PLC and the robot controller are, and give an example. (3)

[20]

QUESTION 6

6.1 Name FOUR items that should be shown on the main HMI screen of the cell. (4)

6.2 In one week the cell had an availability of 90%, a performance of 95% and a quality of 98%. Calculate its OEE. (3)

6.3 Explain what the factory gains by connecting the cell to SCADA. (3)

6.4 Give TWO Industry 4.0 improvements that could be added to the cell. (2)

6.5 Give TWO cybersecurity measures for the connected cell. (2)

6.6 Name ONE communication protocol that could connect the PLC to the HMI. (2)

[16]

QUESTION 7

7.1 Name FIVE safety measures for the palletising cell. (5)

7.2 Explain how muting of the light curtain at the pallet exit works and why it is needed. (3)

7.3 Describe FOUR steps to commission (start up and test) the new cell safely. (4)

7.4 Explain why an operator must never enter the cell while the robot is in automatic mode, and how the cell must be restarted after a guard door has been opened. (4)

[16]

TOTAL SECTION B: [70]

TOTAL: 100

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END OF QUESTION PAPER